

HAOYANG LI

Director of OpenInfra Asia, Linux Foundation



DIGITAL

SOVEREIGNTY

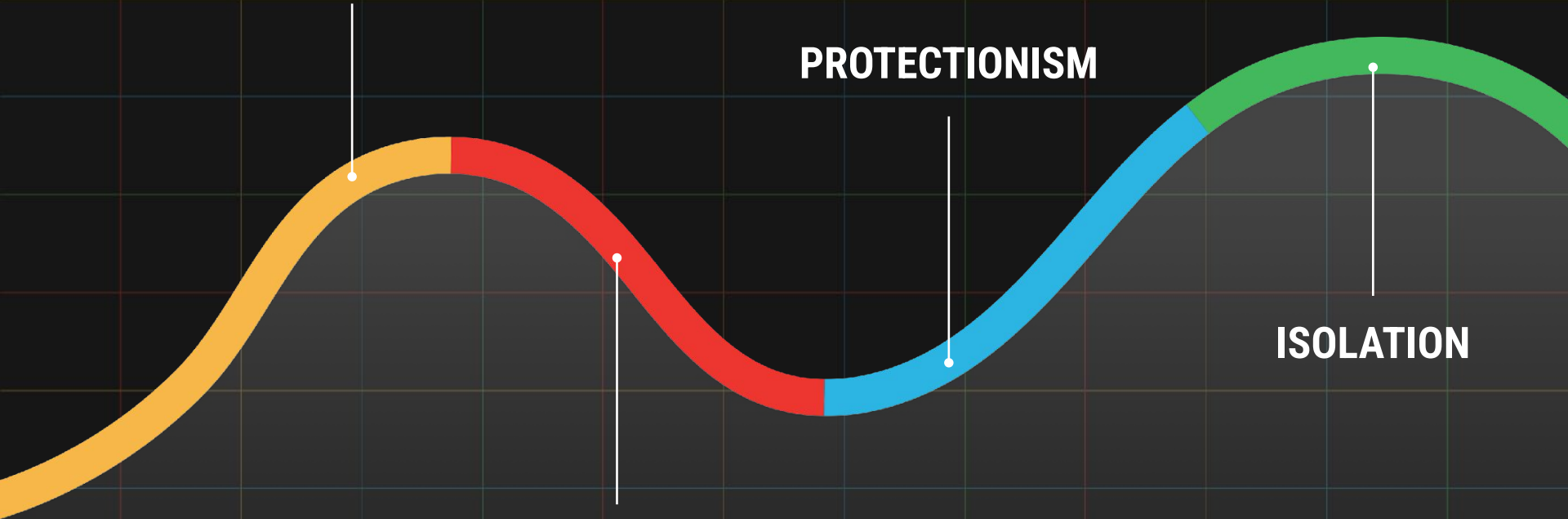


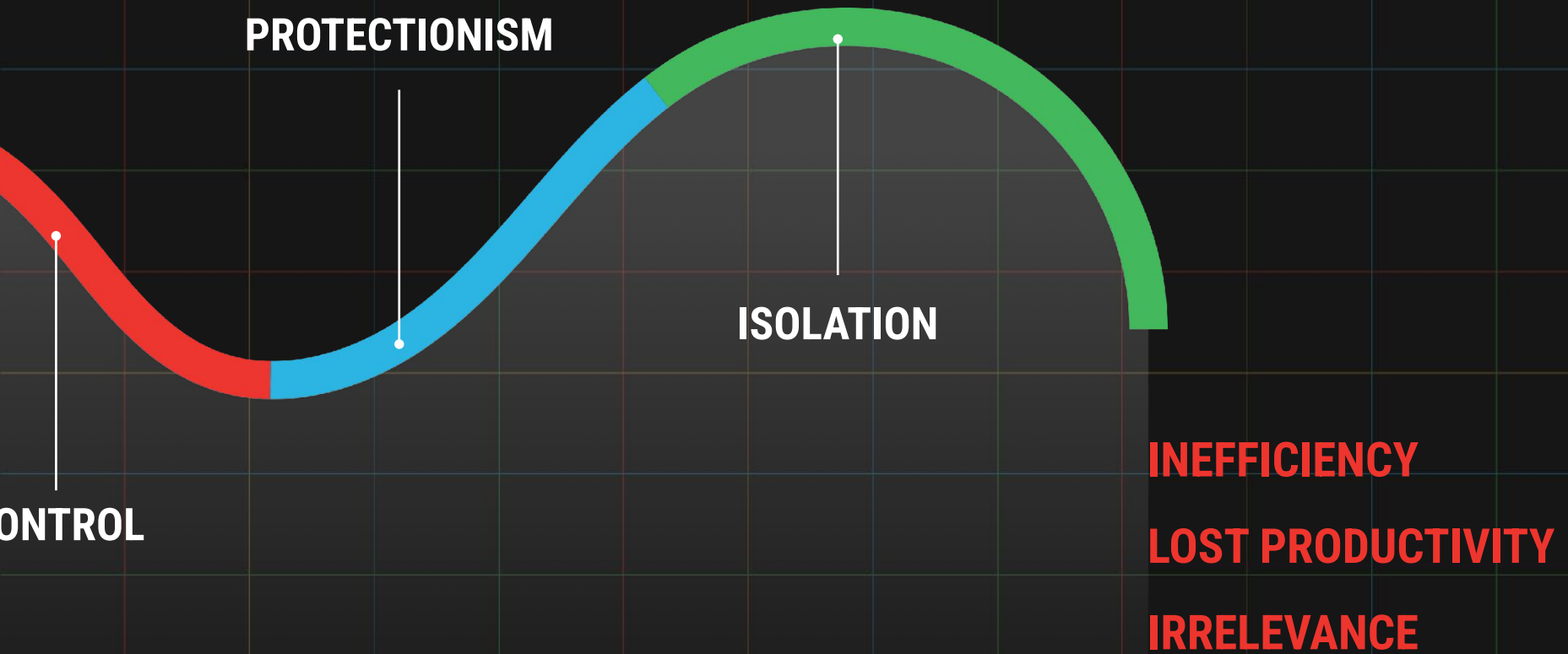
INDEPENDENCE

PROTECTIONISM

ISOLATION

CONTROL







***“OPEN SOURCE ALLOWS
US TO BE SOVEREIGN
WITHOUT BEING
ISOLATIONIST”***



TO GEOPOLITICAL UNCERTAINTY

RESILIENCE

TO CORPORATE CHANGES

TO THREATS & BAD ACTORS



THE WORLD RUNS ON



enedis



UNIVERSITY OF
CAMBRIDGE

MIT

ECMWF

INRAE



KDDI

OVHcloud



GEICO

Alipay

Bloomberg



中国铁路总公司
CHINA RAILWAY



Walmart



LY



kakao



SAP



HYUNDAI



OPENINFRA



→ OPEN INFRASTRUCTURE LEADS THE WAY

OPENINFRA BLUEPRINT IS AN INDUSTRY STANDARD



Linux



openstack.



kubernetes

> The open infrastructure blueprint—built on **Linux, OpenStack, and Kubernetes**—is rapidly gaining adoption across organizations.

> Collaboration among **Linux Foundation, OpenInfra, and CNCF** communities will be key to advancing this combination and accelerating OSS progress.



OpenStack

Kendall Nelson, Linux Foundation





Open Infrastructure for AI: OpenStack's Role in the Next Generation Cloud

OPENSTACK SUPPORT FOR:

- ACCELERATED COMPUTE
- GPU ENABLEMENT
- STORAGE
- NETWORKING

REFERENCE ARCHITECTURES

CASE STUDIES



Kata Containers

Fupan Li, Ant Group



Kata Containers Project Updates

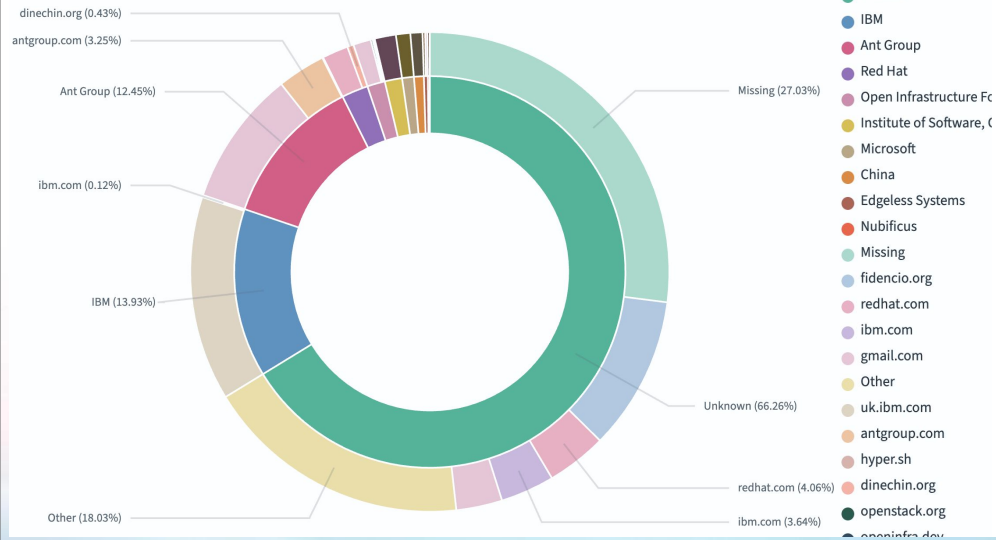
396

Authors

22,056

Contributions

Contributions by Organization and Domains

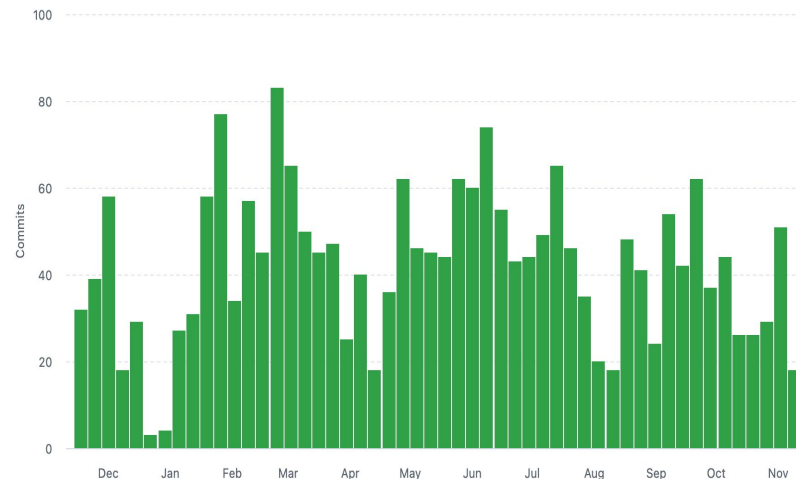


Commits over the last

year

Commits

Number of commits per week



Kata Containers Project Updates

- GPU Passthrough Support with DP & CDI, including hot plug & cold plug
- CoCo (Confidential Container) Support improvements (esp. in runtime-rs)
- Kata runtime-rs improvement (single binary for sandbox + shim), and Kata v4.0 is on its way;
- Nydus v6 Image Support (container image pull acceleration via lazy-load) Related Technologies
- PVM: run Kata Containers without hardware VT



Open Collaboration for Secure AI Workloads - The Kata Containers and Agent Sandbox Integration

By [Zvonko Kaiser](#) on 11/11/2025

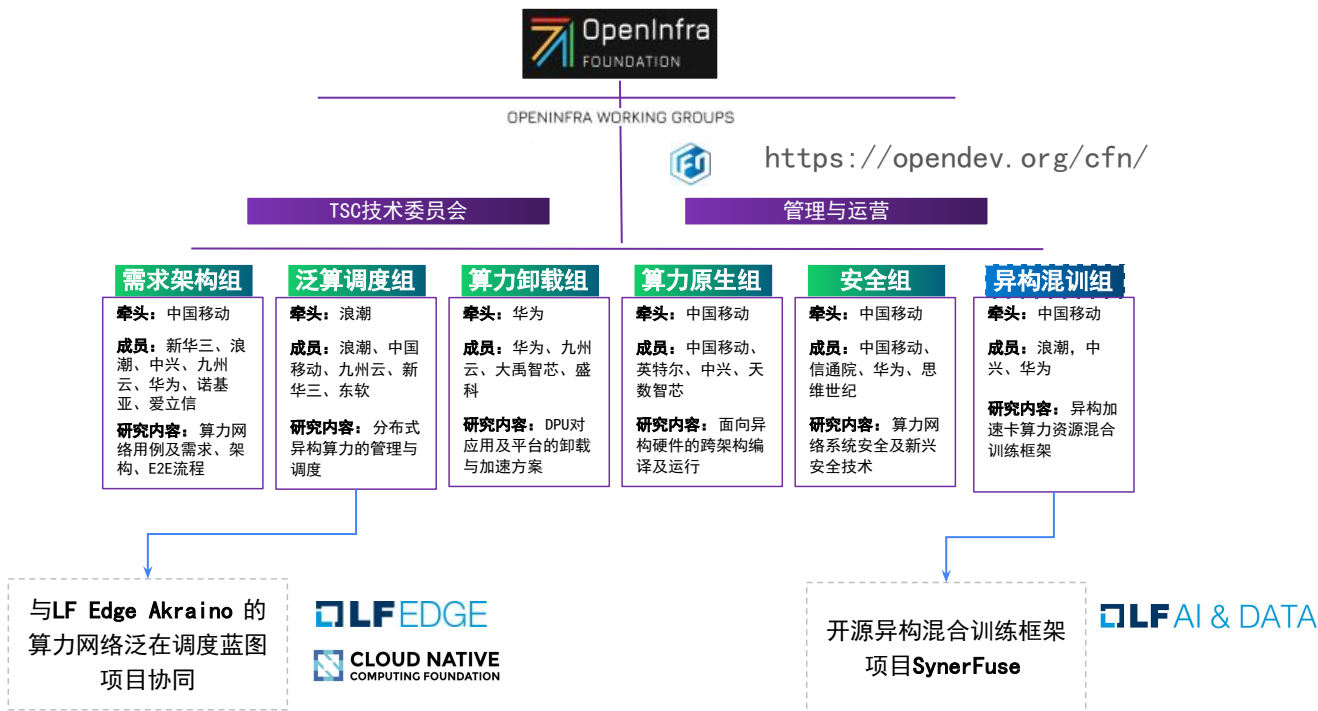
CFN & SynerFuse

Yanjun Chen, China Mobile



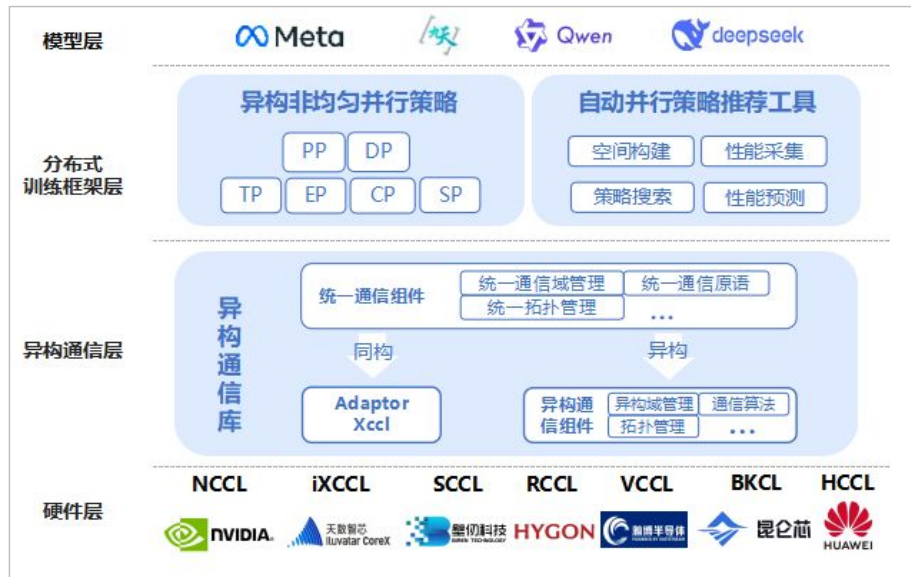
OpenInfra CFN WG算力网络工作组

- 2022年**算力网络工作组**（CFN WG）在OpenInfra社区成立，致力于推动算力网络关键技术的开源实现与代码输出，促进构建开放的算力网络技术生态，并设立需求架构、泛算调度、算力卸载、算力原生、安全、异构混训等6个**子工作推进组**，目前已发布多项Release，入选《央国企开源项目典型实践案例》。



工作组由运营商、主流云厂商、设备商等在内的**19+家成员单位**组成

- 伴随数据样本及模型参数规模的不断增加，为提高智算资源利用率，聚合异构训练算力打破“资源墙”限制，中国移动原创提出异构混训技术并推动在CFN开源，后续计划协同业界热门开源社区进行优势互补，多角度构建全方位混训开源生态体系。



SynerFuse

<https://github.com/SynerFuse/SynerFuse>

核心能力:

- 分布式计算任务非均匀拆解
- 并行策略性能预测生成
- 异构芯片间统一通信库

后续计划:

- 异构多维并行加速、显存优化、重计算、故障定位容错技术
- 探索混训应用案例，形成行业最佳实践



BUILD THE NEXT DECADE OF OPEN INFRASTRUCTURE WITH US!

JOIN US AT



openinfra.org/join



CNCF Today: A global open source community

In the past 10 years,

290,000+ contributors

made

3,350,000+ code commits

1,200,000+ pull requests

and

18,800,000+ contributions

across 230+ projects

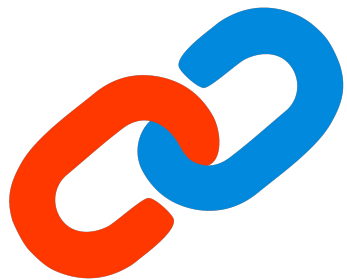
and

190+ countries

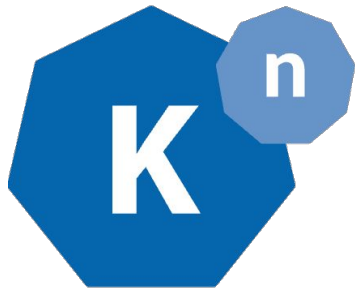
Kubernetes trails Linux (34 years old) as the highest velocity project
github.com/cncf/velocity



Newly graduated projects for 2025



in-toto



Knative



Crossplane



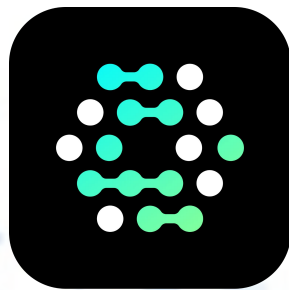
Dragonfly



Newly incubating projects for 2025



Kubescape



OpenFGA™



OpenYurt



Lima



#Projectversaries - Cheers to 10 years



Open Policy Agent



... and many more!



11 years!



11 years!



11 years!



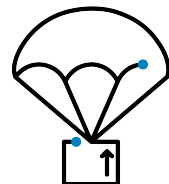
12 years!



14 years!



15 years!



16 years!





**What do the next 10 years
look like?**



The two most significant trends in technology—Cloud Native and AI—are merging



3 Pillars of Open Source AI Today

Agents are getting hyped, but we can't have agents without models and a massive footprint of inference computing

Market Trends
Key Projects



Training

Model Development

PyTorch, DeepSpeed, Volcano,
kubeflow... (dozens more)

PyTorch has **80% share** of
training on Huggingface



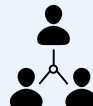
Inference

Model serving

vllm, ONNX Runtime, Kserve,
LMCache, Triton Inference, AlbBrix,
llm-d, SGLang, (dozens more)

1.3 quadrillion tokens/month
(google)

Cost to serve down **33X/year**



Agents

*Connect models to humans, tools,
apps, data, other agents*

MCP
Agent2Agent
Agent Gateway
(MANY more every day)

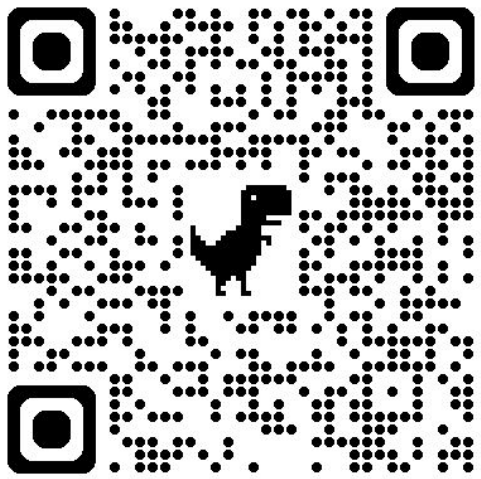
<5%
Production Agent Adoption





Conformance with Kubernetes and AI

Making AI workloads portable and interoperable on Kubernetes



Try It Yourself!

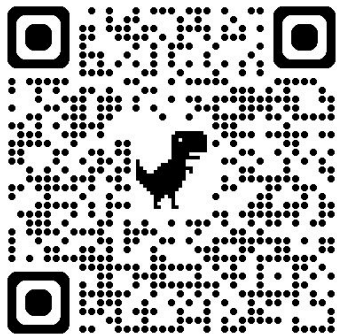


github.com/janetkuo/kubecon-demo

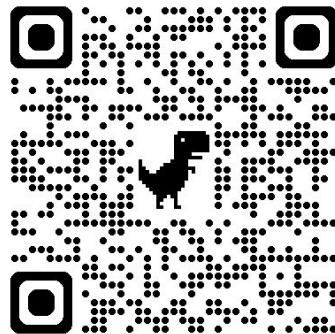
Initial adopters and major contributors



Help us build the Kubernetes Conformance for AI



[cncf/k8s-ai-conformance](https://github.com/cncf/k8s-ai-conformance)



[AI Conformance WG](#)

Get Involved: code, collaborate and advocate

Code Contributions

Technical Oversight Committee

- Technical Advisory Groups (TAGs)
- Working Groups
- CLOTributor.dev
- 230+ Projects and Growing!

Non Code Contributions

Cloud Native Advocacy

- Cartografos WG
- cncf.io/blog
- kubernetes.io/blog
- Cloud Native Glossary
- Project Documentation



See you in 2026!



KubeCon



CloudNativeCon

Europe 2026

Amsterdam | 23-26 March



KubeCon



CloudNativeCon

India 2026

Mumbai | 18-19 June



KubeCon



CloudNativeCon

Japan 2026

Yokohama | 29-30 July



KubeCon



CloudNativeCon

**OpenInfra
SUMMIT > ASIA**

China 2026

Shanghai | 8-9 September



KubeCon



CloudNativeCon

North America 2026

Salt Lake City | November 9-12



KubeCon/CloudNativeCon + OpenInfra Summit Asia 2026



<https://www.lfasiallc.com/kubecon-cloudnativecon-openinfra-summit-china/>



THANK YOU!